



Course Information

Course Code:	MATH 018	Course Name	Linear Algebra with Simulation Software
Units	3	Prerequisite	MATH 105 or equiv.
Course Description	The course is an introduction to the theory of vectors, matrices and linear transformations through the use of computing softwares. The course covers vectors and matrix operations, solvability of linear systems, determinants, dimension of a space, orthogonality, linear transformations, similarity, eigenvalues and eigenvectors, and selected applications.		

Lecturer

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Course Plan

Week	Topic / Activity
1 – 2	Course Orientation
	Vectors and Matrices
3 – 5	Linear Systems
	Nonsingular Matrices
	Determinants
6	Review and Long Examination 1 mode: <i>in-person</i>
7 – 9	Subspaces and Linear Independence
	Basis and Dimension
	Orthonormal Bases
10 – 11	Linear Transformations and Matrix Representations
	Kernel and Range
12	Review and Long Examination 2 mode: <i>in-person</i>
13 – 14	Fundamental Spaces of a Matrix
	The Projection Theorem
15 – 16	Eigenvalues and Eigenvectors
	Diagonalizability
	Spectral Theorem
17 – 18	Review and Long Examination 3 mode: <i>online</i>
	Final Examination mode: <i>in-person</i>

Grading System

(50 – base)

Rating Transmutation Function:

$$\text{Rating} = \left(\frac{\text{Raw Score}}{\text{Max Score}} \times 50 \right) + 50$$

Final Grade Components:

- Class Standing [CS]:
 - long exam 1 – 35%
 - long exam 2 – 35%
 - long exam 3 – 30%
- Major Exam [ME]:
 - final exam

Final Grade [FG]:

$$\text{FG} = (0.7 \times \text{CS}) + (0.3 \times \text{ME})$$

Note: FG is rounded off to the nearest integer and is converted to the University's Point Grade System

Point Grade System

Final Grade	Point Grade
97 – 100	1.00
94 – 96	1.25
91 – 93	1.50
88 – 90	1.75
85 – 87	2.00
82 – 84	2.25
79 – 81	2.50
76 – 78	2.75
75	3.00
50 – 74	5.00
Withdrawn	W
Dropped	D
Incomplete	INC

References and Other Materials

- Linear Algebra**, Jim Hefferon, 4th Edition.
This is a free e-book. Download through this [website](#).
- Elementary Linear Algebra**, Ron Larson, 8th Edition.
- Linear Algebra and Matrix Theory**, Instructional Material, John Patrick Sta. Maria



Class Policies

1. Active Participation

Attend and actively participate on every lecture. MATH 203 is highly cumulative. Don't make it anymore stressful by not exerting an effort. If one misses a lecture, you will like struggle with the future lessons.

2. Focus

Avoid distractions during class lectures. Minimize the use of electronic gadgets to note taking and computing purposes only.

3. Attendance

Attendance will be monitored, but not graded.

4. Absence

Generally, you are allowed to have "bad days" on other days except on exam days. However, an absence during an exam requires **formal excuse**, provided that the reason is valid as deemed valid by the lecturer. Exams are typically planned and announced.

- For **medical reasons**, secure medical certifications from healthcare professionals and have them validated/ authenticated by the University Health Services (UHS) before going to the Office of Counseling and Psychological Services (OCPS), who will issue the absence/admission slip.
- For **non-medical reasons**, bring formal excuse letter to the (OCPS) with the necessary attachments for validation purposes.

5. Missed Long Exam

In case of absence with formal excuse, re-taking the exam is not necessary. The rating for the missed long exam is set to be the minimum rating of the other two long exams.

6. Missed Final Exam

As a short answer, you can't. However, if you must, an INC grade will be encoded in the SIS as your grade. Completion of grades can be made in the regular semester that follows. **Note:** An INC grade lapses to a final grade of 5.00 after a year of not completing the grade.

7. Office Hours

I am usually available for queries and consultation on weekdays, from 8:00AM to 5:00PM. Practice professionalism. Use official channels for communication and use appropriate language, but do not overdo it by sending AI - generated templates.

8. Class Representatives

They are not personal secretaries. Personal concerns must be directed to the lecturer by the student.

9. Academic Integrity

In higher mathematics, collaboration is important. To balance learning with honesty, we impose the following rules.

- **Discuss.** You are encouraged to brainstorm topics and problems, except during an exam.
- **Write.** Once you move from discussion to writing your final submission, you must do so alone.
- **Attribute.** If you worked significantly with a classmate, simply note their name in your work, if applicable.

10. Technology and AI Policy

Using large language models (AI) to generate answers to assessments is a violation of PUP's policy on academic integrity. While the use of scientific calculators and mathematical softwares (like GeoGebra, etc) is encouraged, generative AI kills your growth as a student. Learning to spot the flaws of an AI model can be a good training, but submitting them as your own is a failure of your education.